

(a) Describe how monoclonal antibodies are produced using the hybridoma method.

[5]

Abciximab prevents blood clotting by stopping structures called platelets from binding together.

Fig. 6.1 shows platelets and abciximab in a coronary artery.





State the letter in Fig. 6.1 that represents:

- the target antigen for abciximab
- the constant region of abciximab
- one of the antigen binding sites of abciximab

[3]

(c) Coronary arteries supply oxygenated blood to the cells of the heart.

The passage outlines the cardiac cycle of the heart and the structures in the heart that control the cycle.

Complete the passage by using the most appropriate scientific terms.

The sinoatrial node is located in the wall of the,
one of four chambers of the mammalian heart. Electrical impulses from the sinoatrial node
reach the atrioventricular node, which transmits impulses towards the apex of the heart along
a series of specialised muscle fibres called

Pressure increases in the ventricles when they contract in the stage of the cardiac cycle
known as

[3]

[Total: 11]